

AI/ML Internship – Day 46 Notes & Tasks

Module 6: Understanding YOLO Detection Results, Bounding Boxes, Classes & Confidence Scores

Part 1: Recap of Day 45

Yesterday we learned:

-  Installing YOLOv8
-  Loading Pretrained Models
-  Running First Object Detection
-  Detecting Objects in Images

Today we will understand the detection results generated by YOLO.

Part 2: What Happens After Detection?

Suppose we give this image:

Image.jpg

Image contains:

- Person
- Dog
- Car

YOLO Output:

Person 96%

Dog 93%

Car 91%

But YOLO also provides:

- Object Name (Class)
- Confidence Score
- Bounding Box Coordinates

📌 Part 3: Understanding Classes

◆ What is a Class?

A class is the category of an object.

Examples:

Object Class

Dog Dog

Cat Cat

Car Car

Person Person

Bus Bus

Example

Person → Class

Dog → Class

Car → Class

YOLO predicts these classes automatically.

📌 Part 4: What is a Bounding Box?

A Bounding Box is a rectangle drawn around a detected object.

Example:

```
-----  
|   DOG   |  
-----
```

Purpose:

- ✅ Show object location
- ✅ Highlight detected object

📌 Part 5: Bounding Box Visualization

```
+-----+  
|           |  
|  PERSON   |  
|           |  
+-----+
```

```
+-----+  
|  DOG      |  
+-----+
```

```
+-----+  
|  CAR      |  
+-----+
```

YOLO draws these boxes automatically.

📌 Part 6: Bounding Box Coordinates

Every box has coordinates.

Example:

x = 120

y = 80

width = 200

height = 300

Meaning:

Start Position → (120,80)

Box Width → 200

Box Height → 300

📌 Part 7: Understanding Confidence Score

What is Confidence Score?

Confidence score tells how sure YOLO is about its prediction.

Example

Dog → 98%

Meaning:

YOLO is 98% confident the object is a dog.

More Examples

Prediction Confidence

Dog 98%

Car 94%

Person 89%

Cat 76%

Part 8: High vs Low Confidence

High Confidence

Dog → 99%

Very reliable prediction.

Low Confidence

Dog → 40%

Less reliable prediction.

Part 9: Confidence Threshold

YOLO ignores low-confidence detections.

Example:

```
model.predict(  
    "image.jpg",  
    conf=0.5  
)
```

Meaning

0.5 = 50%

Only detections above 50% confidence are shown.

Part 10: Understanding Detection Output

Example:

image.jpg

Person 0.96

Dog 0.93

Car 0.91

Interpretation:

Person → 96%

Dog → 93%

Car → 91%

All detections are reliable.

Part 11: Accessing Detection Results

Code:

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model("image.jpg")
```

Part 12: Display Detected Classes

```
for result in results:  
    print(result.names)
```

Example Output:

0 : person

1 : bicycle

2 : car

3 : motorcycle

Part 13: Display Bounding Boxes

```
for result in results:  
    print(result.bboxes)
```

Output contains:

Coordinates

Confidence

Class ID

Part 14: Print Confidence Scores

```
for result in results:  
    for box in result.bboxes:  
        print(box.conf)
```

Example:

0.96

0.91

0.88

Part 15: Complete Example Program

```
from ultralytics import YOLO
```

```
model = YOLO("yolov8n.pt")
```

```
results = model("image.jpg")
```

```
for result in results:
```

```
    print("Detected Objects")
```

```
    for box in result.boxes:
```

```
        print("Confidence:",  
              box.conf)
```

Part 16: Real-World Example

Traffic Camera

Image contains:

10 Cars

5 Bikes

2 Buses

15 People

YOLO detects:

Car → 95%

Bike → 93%

Bus → 97%

Person → 96%

Used in:

- Smart Traffic Systems
 - Vehicle Counting
 - Road Monitoring
-

Part 17: Common Mistakes

Mistake 1

Trusting very low confidence predictions.

Example:

Dog → 21%

May be incorrect.

Mistake 2

Ignoring confidence threshold.

Always use:

conf=0.5

or

conf=0.6

Part 18: Theory Questions

What is a Class?

A category assigned to an object.

What is a Bounding Box?

Rectangle around a detected object.

What is Confidence Score?

Confidence level of prediction.

Why use Confidence Threshold?

To remove weak detections.

What information does YOLO provide?

- Object Name
 - Bounding Box
 - Confidence Score
-

Part 19: YOLO Detection Workflow

Image



YOLO Model



Object Detection



Class Identification



Confidence Score



Bounding Box



Final Output

Tasks for Day 46

Theory

1. What is a class?
2. What is a bounding box?
3. What is confidence score?

4. What is confidence threshold?
 5. Why are bounding boxes important?
-

Practical Task 1

Run object detection on any image.

Practical Task 2

Write down:

- Detected Objects
 - Confidence Scores
-

Practical Task 3

Experiment with:

conf=0.3

conf=0.5

conf=0.8

Observe differences.

Practical Task 4

Print:

result.bboxes

and analyze output.

Practical Task 5

Identify:

- Object Class
- Confidence

- Bounding Box

for every detected object.